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EXPERIMENT NO. 07

Aim: To implement monitoring using Prometheus / Grafana / Nagios

LO Mapping: LO5

LO5: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

Theory:

Continuous monitoring is a process to detect, report, respond all the attacks which occur in its infrastructure. Once the application is deployed into the server, the role of continuous monitoring comes into play. The entire process is all about taking care of the company's infrastructure and respond appropriately.

Why We Need Nagios tool?

Here, are the important reasons to use Nagios monitoring tool:

- Detects all types of network or server issues
- Helps you to find the root cause of the problem which allows you to get the permanent solution to the problem
- Active monitoring of your entire infrastructure and business processes
- Allows you to monitor and troubleshoot server performance issues
- Helps you to plan for infrastructure upgrades before outdated systems create failures
- You can maintain the security and availability of the service
- Automatically fix problems in a panic situation
- Nagios is the most popular, open source, powerful monitoring system for any kind of infrastructure. It enables organizations to identify and resolve IT infrastructure problems before they affect critical business processes. Nagios has the capability of monitoring applications, services and entire IT infrastructure.
- NRPE is known as Nagios Remote Plugin Executor. It is designed to execute plugins on remote systems. On Windows hosts, the equivalent remote agent is NSClient++, which listens for Nagios requests and executes local checks (disk, CPU, memory, services, etc.).

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- NSClient++ runs as a Windows service on the remote host and waits for requests from the Nagios server (usually over the NRPE protocol, which NSClient++ also supports). When the Nagios server needs to check a resource on that host, it sends a command; NSClient++ executes the check locally and returns the result to Nagios.

Note on running Nagios on Windows

Nagios Core is built for Linux and does not install natively on Windows. The standard practice for a Windows machine is to run the Nagios Core server inside a Docker container (using Docker Desktop with the WSL2 backend), and to monitor Windows hosts using NSClient++, which is the Windows counterpart of NRPE. All commands below are Windows PowerShell / Command Prompt commands.

1 - Pre-requisite: Install Docker Desktop

Download and install Docker Desktop for Windows and enable the WSL2 based engine. Verify the installation from PowerShell:

```
PS C:\Users\vishal> docker --version
PS C:\Users\vishal> docker version
```

Also make sure the WSL2 feature is enabled:

```
PS C:\Users\vishal> wsl --install
PS C:\Users\vishal> wsl --set-default-version 2
```

2 - Pull and Run the Nagios Core Docker Image

Pull a ready-made Nagios Core image and run it as a container, mapping the web port to the host.

```
PS C:\Users\vishal> docker pull jasonrivers/nagios:latest
PS C:\Users\vishal> docker run --name nagios4 -d -p 8080:80 jasonrivers/nagios:latest
```

Verify the container is up and running:

```
PS C:\Users\vishal> docker ps
```

3 - Set the Apache/Nagios Admin Password

Set (or reset) the password for the nagiosadmin user inside the running container:

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```
PS C:\Users\vishal> docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin
```

Enter and confirm the password when prompted (e.g. 123456).

4 - Access Nagios Web Interface

Access the Nagios web interface from a browser on the Windows host:

```
http://localhost:8080/nagios/
```

Prompting for Authentication Password - **username:**

nagiosadmin

Password: 123456 (which you entered while configuration)

5 - Install NSClient++ on the Windows Remote Host

On the Windows machine you want to monitor, download and silently install NSClient++ (the Windows equivalent of the NRPE agent):

```
PS C:\Users\vishal> Invoke-WebRequest -Uri "https://github.com/mickem/nscp/releases/latest/download/NSCP-0.5.2-x64.msi" -OutFile "NSCP.msi"
PS C:\Users\vishal> Start-Process msiexec.exe -ArgumentList '/i NSCP.msi ADDLOCAL=ALL MODULES=CheckExternalScripts,NRPEServer /qn' -Wait
```

6 - Configure NSClient++

Edit the NSClient++ configuration file and add the Nagios server's IP address to the allowed hosts list:

```
PS C:\Program Files\NSClient++> notepad nsclient.ini allowed hosts=127.0.0.1,192.168.1.100
```

Where 192.168.1.100 is your Nagios server (Docker host) IP address.

After saving the file, restart the NSClient++ service so the change takes effect:

```
PS C:\Users\vishal> Restart-Service nscp
```

7 - Verify Connectivity from Nagios

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From the Nagios server (inside the Docker container), run the NRPE check against the Windows remote host to confirm connectivity. Here 192.168.1.50 is the remote Windows host's IP address.

```
PS C:\Users\vishal> docker exec -it nagios4 /opt/nagios/libexec/check_nrpe -H 192.168.1.50
NRPE v2.15
```

8 - Add the Windows Host in Nagios

Create a configuration file for the new host inside the Nagios container. We also define a service that continuously checks whether the host is reachable (PING check).

```
PS C:\Users\vishal> docker exec -it nagios4 sh -c "vi /opt/nagios/etc/servers/MyWindowsHost001.cfg" define host {
   use             windows-server
host_name       Windows_Host_001
alias           Windows Host 001     address 192.168.1.50
   register        1
}
define service{
   host_name                               Windows_Host_001
service_description           PING              check_command check_ping!100.0,20%!500.0,60%      max_check_attempts 2       check_interval            2       retry_interval            2
check_period              24x7       check_freshness           1
contact_groups            admins       notification_interval   2
notification_period    24x7
   notifications_enabled  1
   register             1
}
```

Verify the configuration file for errors, then restart the Nagios container so the changes take effect:

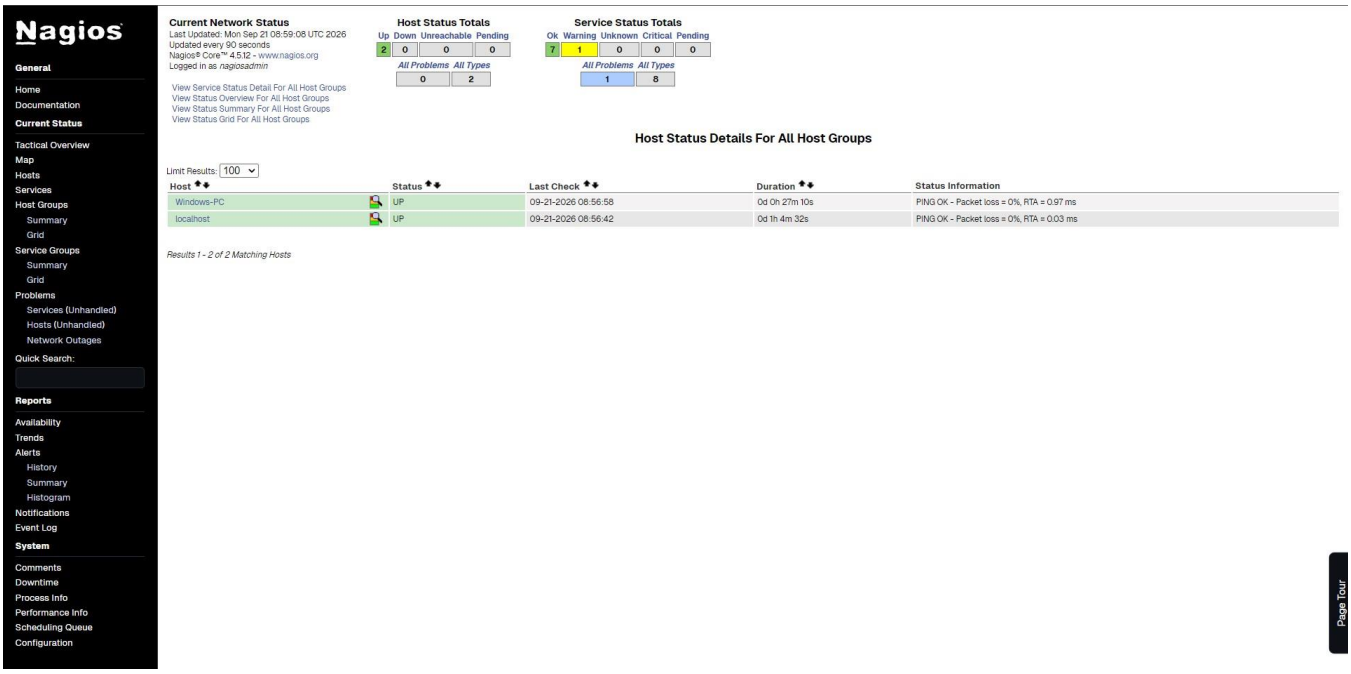
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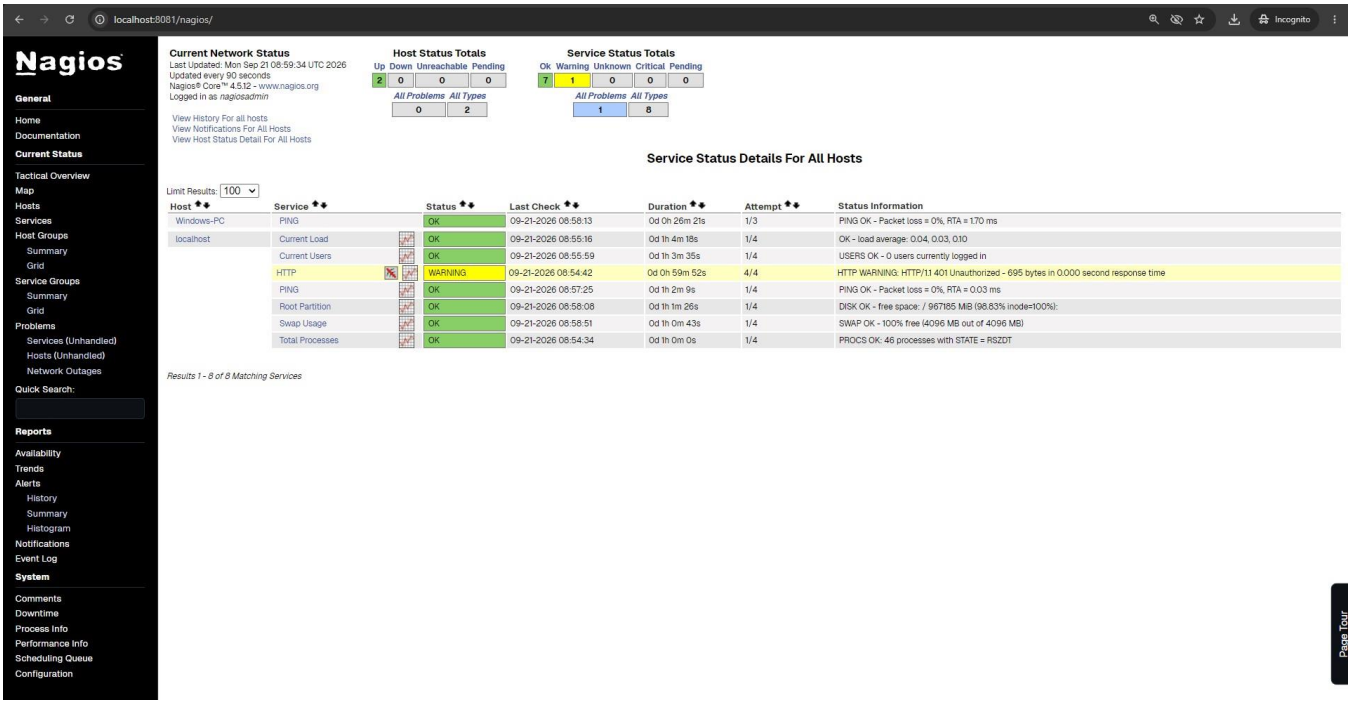
```
PS C:\Users\vishal> docker exec -it nagios4 /opt/nagios/bin/nagios -v
/opt/nagios/etc/nagios.cfg
PS C:\Users\vishal> docker restart nagios4
```

9 - Check Host in Nagios Web Interface

Open the Nagios web interface (http://localhost:8080/nagios/) and check that the new Windows host has been added and is being monitored under Current Status -> Hosts.

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Conclusion:

Nagios Core was successfully installed and configured using Docker to monitor a Windows host. NSClient++ was installed on the Windows system, and the Windows host was successfully added to Nagios. The **PING service** was configured and monitored

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successfully. The host status was displayed as **UP**, with **0% packet loss**, and the PING service status was **OK**. The Nagios configuration was also validated with **0 errors**. Thus,

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the experiment successfully demonstrated continuous monitoring of a Windows system using Nagios.